Framework Specification

A Bayesian Markov Chain Model for Laboratory Biorisk Quantification

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S1. Overview

This document provides a complete mathematical specification of the quantitative biorisk framework introduced in the main manuscript. It is intended as a self-contained reference: a reader with background in applied probability can reproduce all results without consulting the main text.

The framework has five components:

S1. Log-risk metric and companion channel metrics (Section 2)
S2. Five-state Markov chain and edge class taxonomy (Section 3)
S3. Intervention effect functions: training, maintenance, inspection (Section 4)

S4. Bayesian likelihood and prior distributions (Section 5)
S5. Cost-effectiveness metrics (Section 6)

The intervention weight matrix (Section 7) and parameter sensitivity notes (Section 8) complete the specification.

S2. Log-Risk Metric

S2.1. Definition

Definition 1 (Per-procedure log-risk)**.** *Let* cMFPT(S_i) *denote the conditional mean first-passage time to state S_4 (disaster), computed from the forward-escalation sub-chain rooted at S_i after removing probability mass associated with de-escalation below S_i and renormalising rows. The* per-procedure log-risk *rooted at S_i is:*

$$L_i \;:=\; \log_{10}\big(\text{cMFPT}(S_i)\big). \tag{1}$$

Higher L_i means safer operation: $L_i = 7$ implies one expected disaster per 10^7 procedures, $L_i = 6$ implies ten times more risk. This mirrors the decibel convention in acoustics, where a 3 dB increase corresponds to a factor-of-2 change in power.

The restricted sub-chain construction ensures that L_i is sensitive primarily to the intervention class dominating transitions at and above state S_i, enabling clean channel attribution.

S2.2. Channel Metrics

Three channel-specific metrics decompose total risk:

$$L := L_0 = \log_{10}(\text{cMFPT}(S_0)) \qquad \text{(training channel, primary log-risk)} \tag{2}$$

$$L_M := L_1 = \log_{10}(\text{cMFPT}(S_1)) \qquad \text{(maintenance channel)} \tag{3}$$

$$L_I := L_2 = \log_{10}(\text{cMFPT}(S_2)) \qquad \text{(inspection channel)} \tag{4}$$

The triple (L, L_M, L_I) provides non-overlapping ROI attribution: improvements in training shift L without mechanically inflating L_M or L_I, and vice versa.

S2.3. WHO Risk Group Mapping

The log-risk scale maps onto WHO Risk Groups as a calibration anchor (Table 1). This is a modelling choice, not a WHO-endorsed numerical specification.

TABLE 1. Approximate log-risk values by WHO Risk Group at baseline (no intervention improvements above minimum standards).

WHO Risk Group	L (approx.)	P_{disaster} per procedure
RG-1	≈ 2	10^{-2}
RG-2	≈ 4	10^{-4}
RG-3	≈ 6	10^{-6}
RG-4	≈ 8	10^{-8}

S2.4. One-Step Disaster Probability

For reporting purposes, the one-step per-procedure disaster probability is computed directly from the stationary distribution:

$$P_{\text{disaster}} \;=\; \sum_{k=0}^{3} \pi_k \, P_{k4}, \qquad \pi_0 = (1,0,0,0,0)^T. \tag{5}$$

In the typical operating range L and $-\log_{10}(P_{\text{disaster}})$ agree within $\sim$0.1 log units.

S3. Five-State Markov Chain

S3.1. State Space

Laboratory operations are represented as a discrete-time, discrete-state Markov chain $\{X_n\}_{n\geq 0}$ on the state space $\mathcal{S} = \{S_0, S_1, S_2, S_3, S_4\}$:

State	Label	Interpretation
S_0	Normal	All systems functional, procedures on-protocol
S_1	Minor	Procedural deviation or fatigue onset
S_2	Serious	Equipment malfunction or protocol failure
S_3	Critical	Containment threatened or compromised
S_4	Disaster	Laboratory-acquired infection / release (absorbing)

State S_4 is absorbing: $P_{44} = 1$. Each step corresponds to one laboratory procedure.

S3.2. Edge Class Taxonomy

Each non-self transition $(S_i \to S_j)$ carries a *cause-class* tag that links it to the dominant failure mechanism and determines which interventions carry nonzero weight on that edge (Table 2).

Table 2. Edge cause-class taxonomy.

Class	Label	Dominant mechanism / intervention sensitivity
H	Human	Training only
E	Equipment	Maintenance only
HE	Human–Equipment	Training and maintenance (dual class)
EO	Equipment–Org.	Maintenance and inspection (dual class)
–	Uncontrolled	No operational intervention
Natech	Exogenous	Natural/industrial hazard; no operational weight

S3.3. Log-Space Transition Update

Each transition probability is parameterised in log-space. The baseline log-probability for edge (i, j) is $L_{ij}^{\text{base}} = -\log_{10}(P_{ij}^{\text{base}})$. The intervention-adjusted log-probability is:

$$L_{ij}(\theta) \;=\; L_{ij}^{\text{base}} \;+\; w_{ij}^{\text{train}}\,\Delta L_{\text{train}} \;+\; w_{ij}^{\text{maint}}\,\Delta L_{\text{maint}} \;+\; w_{ij}^{\text{insp}}\,\Delta L_{\text{insp}}, \tag{6}$$

where $\theta = (h, f, a, s)$ collects the four intervention parameters (training hours, maintenance frequency, maintenance adherence, inspection score), and $w_{ij}^{(\cdot)} = 0$ whenever the intervention class is inconsistent with the edge cause class. The transition probability is recovered as:

$$P_{ij}(\theta) \;=\; 10^{-L_{ij}(\theta)}. \tag{7}$$

Rows are renormalised after applying interventions to ensure $\sum_j P_{ij} = 1$.

S4. Intervention Effect Functions

S4.1. Training

Training exhibits diminishing returns, modelled by a Hill function:

$$\Delta L_{\text{train}}(h) \;=\; \Delta L_{\text{max}}^{T} \cdot \frac{(h/h_{50})^{n}}{1 + (h/h_{50})^{n}}, \tag{8}$$

with parameters:

Symbol	Value	Interpretation	Basis
$\Delta L_{\text{max}}^{T}$	2.0	Max log-risk improvement	Factor-of-100 expert error reduction
h_{50}	50 h	Half-saturation training hours	≈ one week intensive training
n	2.5	Hill exponent (S-curve steepness)	Empirical learning curve literature

Remark 1. *Results are robust to ±20 % variation in h_{50} or n. The dominant sensitivity is to ΔL^{T}_{max}, which can be updated via the Bayesian pipeline when empirical data are available.*

S4.2. Maintenance

Maintenance effectiveness depends on both scheduled frequency f (sessions per month) and schedule adherence $a \in [0, 1]$ (proportion of sessions completed):

$$\Delta L_{\text{maint}}(f, a) = \Delta L_{\text{freq}}(f) \cdot m(a) \cdot g(a), \tag{9}$$

$$\Delta L_{\text{freq}}(f) = \Delta L^{M}_{\text{max}} \cdot \frac{f^{1.5}}{1 + f^{1.5}}, \tag{10}$$

$$m(a) = \min\big(1,\ \alpha\, a\big), \quad \alpha = 1.2, \tag{11}$$

$$g(a) = e^{-\beta(1-a)}, \quad \beta = 3. \tag{12}$$

Symbol	Value	Interpretation	Basis
$\Delta L^{M}_{\text{max}}$	1.2	Max log-risk improvement from maintenance	Equipment failure ∼factor-of-16 reducible
$f_{1/2}$	1.0/mo	Half-saturation frequency	Standard BSL-2/3 maintenance cycle
α	1.2	Adherence synergy factor	Functional maintenance culture bonus
β	3	Gap penalty coefficient	Poisson survival: $k \approx 3$ failures per missed session

The gap penalty $g(a)$ captures accelerating deterioration during missed maintenance windows. Its effect is nonlinear:

Adherence a	Gap penalty $g(a)$
0.95	0.861
0.90	0.741
0.80	0.549
0.70	0.407
0.50	0.223

This nonlinearity is the mathematical basis for the *adherence-dominates-frequency* result reported in the main manuscript.

S4.3. Inspection

Inspection produces a binary threshold effect reflecting pass/fail audit logic:

$$\Delta L_{\text{insp}}(s) \ = \ \begin{cases} 1.0 & \text{if } s \geq 70, \\ 0.0 & \text{otherwise}, \end{cases} \tag{13}$$

where $s \in [0, 100]$ is the inspection score.

The 1.0-log input is distributed through the *EO* weight coefficients (0.6 on both $S_2 \to S_4$ and $S_3 \to S_4$), yielding a net system-level gain:

$$\Delta L_I \ = \ 0.6 \times 1.0 \ = \ 0.600 \text{ logs}, \tag{14}$$

a factor of ∼4 reduction in conditional disaster probability from states S_2 or S_3.

S4.4. Combined Update

Effects are additive in log-space. The total log-space improvement for transition (i, j) is:

$$\Delta L_{ij}(\theta) \ = \ w^{\text{train}}_{ij}\, \Delta L_{\text{train}}(h) + w^{\text{maint}}_{ij}\, \Delta L_{\text{maint}}(f, a) + w^{\text{insp}}_{ij}\, \Delta L_{\text{insp}}(s). \tag{15}$$

Additivity in log-space corresponds to multiplicative independence of intervention effects in probability space — a simplifying assumption discussed in the limitations section of the main text.

S5. Bayesian Likelihood and Priors

S5.1. Observable Event Types

Three event types are observable per period of N procedures:

- **Near-misses** (n): transitions terminating in S_1; rate $p_{01}(\theta) = [\pi_0 \mathbf{P}(\theta)]_1$.
- **Light incidents** (ℓ): $S_2 \to S_3$ escalations; rate $p_{23}(\theta) = P_{23}(\theta)$.
- **Disasters** (d): absorptions in S_4; rate $p_{S_4}(\theta) = [\pi_0 \mathbf{P}(\theta)]_4$.

S5.2. Joint Likelihood

Assuming conditional independence of the three event channels:

$$\mathcal{L}(d, \ell, n \mid \theta) \;=\; \mathrm{Bin}(d \mid N,\, p_{S_4}) \cdot \mathrm{Bin}(\ell \mid N,\, p_{23}) \cdot \mathrm{Bin}(n \mid N,\, p_{01}), \tag{16}$$

where $\theta = (h, f, a, s)$.

Remark 2. *The light-incident channel (ℓ) substantially improves identifiability of maintenance and inspection parameters compared to a disaster-only likelihood. This motivates the three-observable design: near-miss data constrain the training channel; light incidents constrain the maintenance and inspection channels.*

S5.3. Prior Distributions

Table 3. Prior distributions, effective ranges, and literature basis.

Parameter	Prior	95 % prior range	Literature basis
Training h (h)	LogNormal($\ln 50$, 0.8)	18–134 h	Initial 2 h + annual refreshers over career
Maint. freq. f (mo^{-1})	LogNormal($\ln 1.0$, 0.7)	0.37–2.7/mo	Standard BSL-2/3 monthly inspection cycles
Maint. adh. a	Beta(2, 2)	0.09–0.91	40–60 % typical compliance (conservative)
Inspection s	Uniform(0, 100)	0–100	Complete prior ignorance on audit score

S5.4. MCMC Implementation

Posterior sampling uses a Metropolis–Hastings random-walk sampler with:

- 20,000 iterations, burn-in 5,000, thinning factor 5
- Proposal: independent Gaussian steps per parameter, step sizes tuned to target acceptance rate 0.20–0.50
- Convergence: Gelman–Rubin $\hat{R} < 1.1$ on three parallel chains

S6. Cost-Effectiveness Metrics

S6.1. Monetary Benefit

For an intervention that shifts log-risk from L_0 to $L_1 > L_0$:

$$B \;=\; N_{\mathrm{proc}} \cdot V \cdot \left(10^{-L_0} - 10^{-L_1}\right) \quad [\$/\mathrm{yr}], \tag{17}$$

with $N_{\mathrm{proc}} = 10^4$ procedures per year and $V = \$50 \times 10^6$ (combined direct, indirect, and intangible costs of a laboratory disaster; order-of-magnitude estimate).

S6.2. ROI and Payback

$$\text{ROI} := \frac{B}{C} \tag{18}$$

$$\text{Payback} := \frac{C}{B} \quad \text{(years)} \tag{19}$$

where C is the annualised intervention cost.

S6.3. Dimensionless Efficiency

For maintenance and inspection, where monetary conversion is less direct, a dimensionless efficiency metric is used:

$$\eta_M := \frac{\Delta L_M}{C/1000} \quad [\Delta L_M \text{ per \$1k}], \qquad \eta_I := \frac{\Delta L_I}{C/1000} \quad [\Delta L_I \text{ per \$1k}]. \tag{20}$$

S6.4. Assumed Intervention Costs

These are order-of-magnitude estimates adjustable per facility:

Intervention	Unit cost	Basis
Training	\$100 per hour	Instructor + staff time + materials
Maintenance frequency	\$10,000 per 1.0/mo increment	Additional staff, parts, scheduling
Inspection (to pass threshold)	\$30,000 one-time	Consulting, facility upgrades, audit fee

S7. Intervention Weight Matrix

TABLE 4. Complete intervention weight matrix. Base rates are $-\log_{10}(P_{ij}^{\text{base}})$. Zero weights are omitted for clarity.

Transition	Dominant mechanism	Class	Base rate	w^{train}	w^{maint}	w^{insp}
$S_0 \to S_1$	Procedural deviation	H	3.0	0.3	–	–
$S_0 \to S_2$	Spontaneous equip. failure	E	6.5	–	0.8	–
$S_0 \to S_4$	Natech exogenous bypass	Natech	7.5	–	–	–
$S_1 \to S_0$	Procedural recovery	H	2.0	0.5	–	–
$S_1 \to S_2$	Equipment malfunction	E	3.5	–	0.8	–
$S_1 \to S_4$	Deviation to exposure	HE	6.0	0.3	0.5	–
$S_2 \to S_0$	Corrective recovery	E	1.5	–	0.8	–
$S_2 \to S_1$	Partial de-escalation	–	2.0	–	–	–
$S_2 \to S_3$	Technical escalation	E	3.0	–	0.8	–
$S_2 \to S_4$	Technical failure to exposure	EO	5.0	–	0.6	0.6
$S_3 \to S_2$	Partial stabilisation	–	1.0	–	–	–
$S_3 \to S_4$	Containment failure	EO	3.5	–	0.5	0.6

Design rationale.

- **Training weights** act exclusively on human-performance transitions (H/HE edges): deviation onset, recovery, and the deviation-to-exposure shortcut. Training cannot reduce equipment or organisational failure rates.
- **Maintenance weights** target all equipment-dominated edges ($E/HE/EO$): spontaneous failure, escalation steps, corrective recovery, and dual-class exposure paths.

- **Inspection weights** act exclusively on *EO* edges proximate to S_4 ($S_2 \to S_4$, $S_3 \to S_4$), reinforcing the final containment barrier. Inspection does not affect procedural or equipment failure rates earlier in the chain.
- **Uncontrolled edges** ($-$ class) and the **Natech** bypass carry no weight; they are not amenable to the three modelled interventions.

S8. Parameter Sensitivity

TABLE 5. Sensitivity of key outputs to ±20 % parameter variation from reference values. Impact rated Low/Medium/High.

Parameter	Affected output	Direction	Impact
$\Delta L_{\max}^{T}$ (training ceiling)	L, training ROI	Proportional	High
h_{50} (training half-sat.)	Optimal training hours	Shifts saturation	Medium
β (gap penalty coefficient)	Adherence-vs-frequency result	Amplifies gap effect	High
$\Delta L_{\max}^{M}$ (maint. ceiling)	L_M, maint. ROI	Proportional	Medium
Inspection threshold (70)	ΔL_I achievability	Binary shift	Medium
V (disaster cost)	All monetary ROI values	Proportional	High
N_{proc} (annual procedures)	All monetary ROI values	Proportional	High

Remark 3. *The* qualitative *findings (adherence dominates frequency; inspection has highest dimensionless return; training saturates above 80 h) are robust to all parameter variations tested. Only the* quantitative *values (specific ROI multiples, dollar figures) are sensitive to V and N_{proc}, both of which are facility-specific inputs.*

S9. Code Availability

The full Python implementation (v5.4) is available at:
https://github.com/dprodanov/biorisk
The repository includes:

- Markov chain construction and cMFPT computation
- All three intervention effect functions
- MCMC sampler with convergence diagnostics
- ROI and budget optimisation routines
- Reproducible scripts for all tables and figures in the main manuscript